EHC 26 9–12 December 2026 Berlin (DE) and online
20th European Headache Congress
P548

Scan me, for digital PDF version:

References

COMPARISON OF THE EFFICACY BETWEEN DULOXETINE AND AMITRIPTYLINE FOR THE PREVENTIVE TREATMENT OF MIGRAINE

Daniel Neto [1], Arthur Soares [2], Norma Espindola [3], Prof. Dr. Wagner Horta [1,2]

[1] *Faculdade de Medicina de Olinda - FMO, Liga Acadêmica de Neurologia, Olinda, Brazil,* [2] *Universidade de Pernambuco, Faculdade de Ciências Médicas, Recife, Brazil,* [3] *Centro Universitário Mauricio de Nassau, Recife, Brasil*

Background and Aims

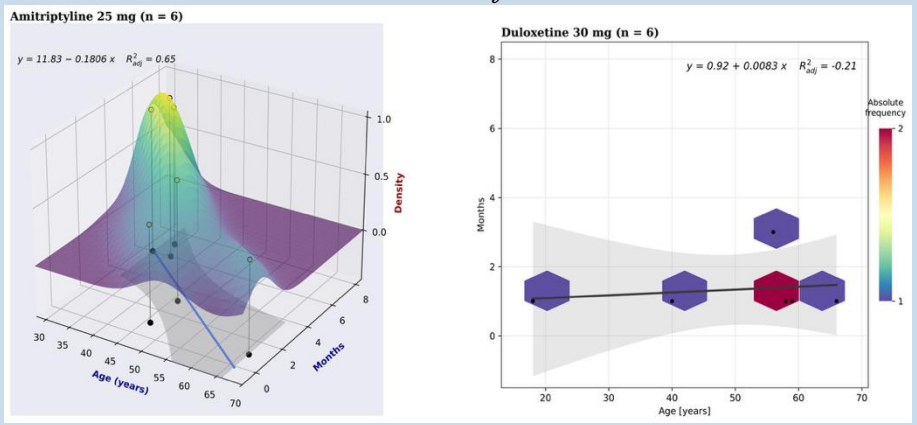
Migraine is a primary headache disorder characterized by generally unilateral, pulsatile pain of moderate to severe intensity, frequently associated with nausea, photophobia, and phonophobia, and may occur with or without aura. It has an estimated prevalence of 10-12% in the population, with higher occurrence in women, and can lead to significant functional impairment and reduced quality of life. Its pathophysiology involves hyperactivation of the trigeminovascular system, alterations in serotonergic neurotransmission, hypersensitivity to sensory stimuli. Diagnosis is clinical and based on the criteria of the International Classification of Headache Disorders. Treatment includes both abortive and preventive therapies, with antidepressants widely used for migraine prophylaxis, particularly duloxetine (DUL) and amitriptyline (AMI). In this context, the presente study aims to compare the efficacy of (DUL) and (AMI) in the preventive treatment of migraine.

Objective

Compair the efficacy between duloxetine and amitriptyline for the preventive treatment of migraine.

Methods

The methodology of this study consisted of a retrospective analysis of data extracted from medical records and follow-up forms of patients diagnosed with chronic migraine or tension-type headache. Data collection focused on variables such as age, sex, prophylactic medication used (AMI × DUL), time to observed crisis reduction, and perceived clinical efficacy. Statistical analysis was performed using Epi Info software. The data were processed to obtain descriptive statistics, including absolute and relative frequencies for categorical variables(sex and efficacy), as well as means and standard deviations for continuous variables(age and response time). Based on the analysis of the results, it is concluded that both (AMI) and (DUL)are highly effective options for the preventive treatment of migraine, achieving a reduction in crisis frequency in all evaluated patients. However, (DUL) demonstrated superiority in the speed of prophylactic action onset, reducing crises in a considerably shorter average time than (AMI). The epidemiological profile reinforces the current literature on the higher prevalence of migraine in females. The choice between the two medications can be guided by the patient's age and the urgency of clinical response, with (DUL) being a promising alternative for a faster reduction of symptoms, especially in the older age groups observed in this study.



Results

A total of 12 female patients were evaluated: six receiving amitriptyline (AMI; 25 mg) and six receiving duloxetine (DUL; 30 mg). Both groups showed a reduction in headache attack frequency, corresponding to a response rate of 100% (6/6) for each medication. In the AMI group, the mean age was 43.3 ± 12.6 years, with a median of 41 years and a range of 32–66 years. The mean time to reduction in headache attack frequency was 4.0 ± 2.7 months, with a median of 4.5 months and a range of 1–7 months. The Shapiro–Wilk test did not indicate a statistically significant departure from normality for response time (W = 0.848; p = 0.152). In the DUL group, the mean age was 49.5 ± 17.7 years, with a median of 57 years and a range of 18–66 years. The mean time to reduction in headache attack frequency was 1.3 ± 0.8 months, with a median of 1 month and a range of 1–3 months. The Shapiro–Wilk test indicated a non-normal distribution of response time (W = 0.496; p < 0.001). Descriptively, the DUL group showed a shorter mean time to reduction in headache attack frequency than the AMI group. However, given the small sample size and the absence of an inferential comparative analysis between treatments, these findings do not establish the therapeutic superiority of either medication.

Conclusions

Based on the analysis of the results, it is concluded that both(AMI)and (DUL)are highly effective options for the preventive treatment of migraine, achieving a reduction in crisis frequency in all evaluated patients.However, (DUL) demonstrated superiority in the speed of prophylactic action onset, reducing crises in a considerably shorter average time than(AMI).The epidemiological profile reinforces the current literature on the higher prevalence of migraine in females.The choice between the two medications can be guided by the patient's age and the urgency of clinical response, with (DUL) being a promising alternative for a faster reduction of symptoms, especially in the older age groups observed in this study.

